

Chapter 4C: Acid-Base Disorders and Integrated Electrolyte Cases

Medvale Internal Medicine - Part V: Nephrology, Fluids, and Acid-Base

How to use this chapter

Acid-base interpretation is a pattern-recognition problem with arithmetic guardrails. The key skill is to name the primary process, verify whether compensation is appropriate, and then search for the mechanism that connects the blood gas to the patient in front of you. This chapter is original Medvale content generated from the renal/fluids source as a coverage signal and cross-checked against current clinical references.

Why this matters clinically

Acid-base disorders are not isolated laboratory abnormalities. They are compressed stories about ventilation, perfusion, kidney function, electrolytes, drugs, toxins, endocrine signals, and tissue metabolism. A patient with sepsis may present with lactic acidosis before blood pressure falls. A patient with vomiting may develop chloride-responsive metabolic alkalosis while simultaneously losing potassium. A patient with COPD can live for years with compensated respiratory acidosis and then abruptly decompensate when pneumonia, sedatives, or excessive oxygen worsens ventilation. A blood gas therefore gives a real-time physiologic map: pH reports danger, PCO₂ reports ventilation, bicarbonate reports metabolic buffer status, and oxygenation must be interpreted separately from acid-base status.

Numerical anchors keep the interpretation safe. Normal arterial pH is approximately 7.35-7.45; PCO₂ is about 35-45 mm Hg; serum bicarbonate is about 22-28 mEq/L. A pH below 7.20 suggests clinically important acidemia, and pH near or below 7.10 should trigger urgent evaluation of perfusion, ventilation, potassium, toxin exposure, renal failure, and need for ICU-level care. The anion gap, compensation formulas, and osmolar gap are not decorations; they prevent false closure by exposing mixed disorders.

Figure 4C-1. Four primary acid-base buckets

Metabolic acidosis pH low, HCO ₃ low Respiratory compensation	Respiratory acidosis pH low, PCO ₂ high Renal compensation
Metabolic alkalosis pH high, HCO ₃ high Respiratory compensation	Respiratory alkalosis pH high, PCO ₂ low Renal compensation

First name the pH direction, then identify the variable moving in the same direction as the pH abnormality.

Core illness scripts

Syndrome	Typical story	Mechanism link	First dangerous question
High anion gap metabolic acidosis	Shock, sepsis, DKA, renal failure, toxic ingestion, seizure, ischemic bowel	New acid consumes bicarbonate; gap rises as unmeasured anions accumulate	Is there impaired perfusion, ketoacidosis, renal failure, salicylate, methanol, or ethylene glycol?
Normal anion gap metabolic acidosis	Diarrhea, ileostomy, renal tubular acidosis, acetazolamide, large saline load	Bicarbonate is lost or acid excretion is impaired; chloride rises to preserve electroneutrality	Is the kidney appropriately excreting ammonium, or is this extrarenal bicarbonate loss?

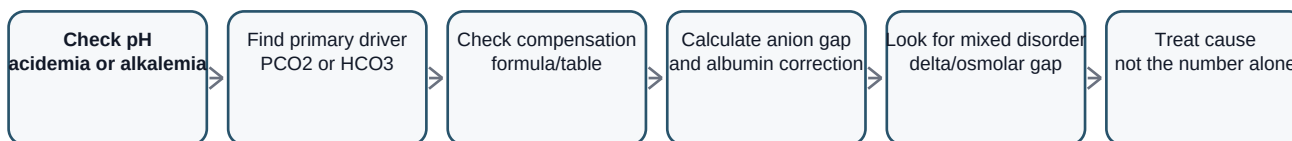
Syndrome	Typical story	Mechanism link	First dangerous question
Metabolic alkalosis	Vomiting, NG suction, loop/thiazide diuretics, mineralocorticoid excess, post-hypercapnia	Hydrogen/chloride/potassium losses plus kidney maintenance of high bicarbonate	Is urine chloride low, suggesting chloride-responsive alkalosis?
Respiratory acidosis	COPD, obesity hypoventilation, sedatives/opioids, neuromuscular weakness, severe asthma	Alveolar hypoventilation retains CO ₂ ; kidney raises bicarbonate if chronic	Is the patient failing ventilation and needing noninvasive or invasive support?
Respiratory alkalosis	Anxiety/pain, pregnancy, hypoxemia, PE, sepsis, liver disease, salicylates	Alveolar hyperventilation lowers CO ₂ ; kidney lowers bicarbonate if chronic	Is this early sepsis, PE, hypoxemia, or salicylate toxicity rather than anxiety?

Pathophysiology: buffers, lungs, and kidneys

The extracellular bicarbonate buffer system links carbon dioxide and bicarbonate through carbonic acid. The lungs regulate the volatile acid side of the system by changing alveolar ventilation. The kidneys regulate the nonvolatile acid side by reclaiming filtered bicarbonate, generating new bicarbonate, secreting hydrogen ions, and excreting ammonium and titratable acid. Compensation never fully normalizes pH in an uncomplicated disorder. If pH is normal but PCO₂ and bicarbonate are both abnormal, a mixed disorder is likely rather than a solved problem.

Respiratory compensation occurs within minutes because ventilation can change quickly. Renal compensation takes hours to days because it depends on tubular transport, ammoniogenesis, and changes in bicarbonate handling. This timing distinction is the basis for acute versus chronic respiratory formulas. A COPD patient with chronic CO₂ retention may have a bicarbonate in the mid-30s and a near-normal pH; the same PCO₂ in an opioid overdose patient without renal compensation can produce severe acidemia.

Figure 4C-3. The six-step blood gas workflow



ABG and VBG interpretation

An arterial blood gas is preferred when oxygenation, alveolar-arterial gradient, carbon monoxide poisoning, severe shock, or exact PaO₂ matters. A venous blood gas is often adequate for acid-base screening in stable patients because venous pH is usually slightly lower and venous PCO₂ slightly higher than arterial values, but VBG cannot replace ABG for oxygenation decisions. Pulse oximetry reports hemoglobin saturation, not ventilation; a patient can have normal SpO₂ and dangerous hypercapnia.

Start by checking whether the sample fits the patient. A dramatic blood gas should match the bedside story. A low PCO₂ in an obtunded hypoventilating patient, a high oxygen tension in a patient not receiving oxygen, or a normal pH in a critically ill patient with extreme bicarbonate values should prompt repeat testing and review of sample source.

Step	Question	Interpretation rule
1. pH	Is pH below 7.35 or above 7.45?	Low pH = acidemia. High pH = alkalemia. Normal pH does not exclude a mixed disorder.
2. PCO ₂	Does PCO ₂ move opposite the pH?	High PCO ₂ lowers pH, so it points toward respiratory acidosis. Low PCO ₂ raises pH, so it points toward respiratory alkalosis.
3. HCO ₃	Does bicarbonate move with the pH?	Low bicarbonate lowers pH, so it points toward metabolic acidosis. High bicarbonate raises pH, so it points toward metabolic alkalosis.

Step	Question	Interpretation rule
4. Compensation	Is the compensatory variable close to expected?	Too little or too much compensation means there is an additional primary process.
5. Gaps	Is the anion gap or osmolar gap elevated?	Gaps reveal unmeasured anions or osmotically active solutes.
6. Context	What illness explains the math?	No formula diagnoses a patient by itself. Link the result to perfusion, ventilation, kidneys, drugs, and toxins.

Compensation formulas

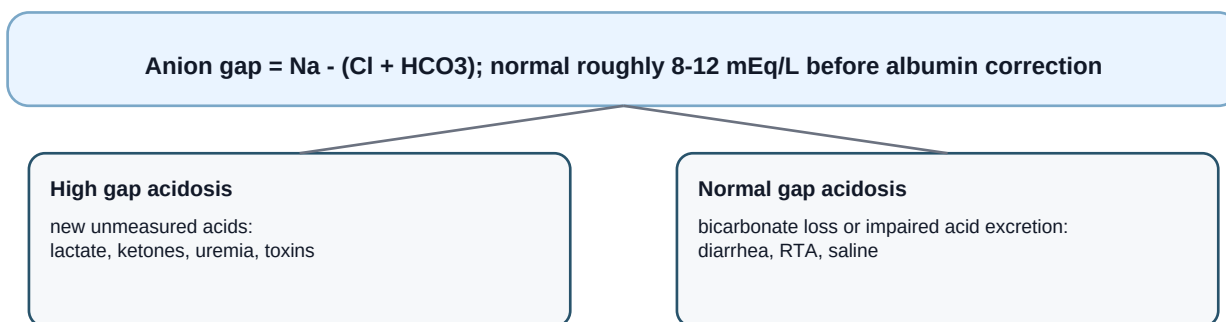
Compensation formulas are error detectors. They do not prove the diagnosis, but they show whether the body response is proportional to the primary disorder. Compensation should move pH toward normal but not cross normal into the opposite pH direction. When the measured value falls outside the expected range, name the added process and explain it clinically.

Primary disorder	Expected compensation	Interpretation of mismatch
Metabolic acidosis	Expected PCO ₂ = 1.5 x HCO ₃ + 8 +/- 2	Measured PCO ₂ higher = added respiratory acidosis. Lower = added respiratory alkalosis.
Metabolic alkalosis	Expected PCO ₂ = 40 + 0.7 x (HCO ₃ - 24) +/- 5	Measured PCO ₂ lower = added respiratory alkalosis. Higher = added respiratory acidosis.
Acute respiratory acidosis	HCO ₃ rises about 1 mEq/L for each 10 mm Hg PCO ₂ above 40	Bigger bicarbonate rise suggests chronicity or metabolic alkalosis.
Chronic respiratory acidosis	HCO ₃ rises about 3.5-4 mEq/L for each 10 mm Hg PCO ₂ above 40	Smaller rise suggests acute-on-chronic hypercapnia or metabolic acidosis.
Acute respiratory alkalosis	HCO ₃ falls about 2 mEq/L for each 10 mm Hg PCO ₂ below 40	Large fall suggests added metabolic acidosis.
Chronic respiratory alkalosis	HCO ₃ falls about 4-5 mEq/L for each 10 mm Hg PCO ₂ below 40	Small fall suggests acute process or metabolic alkalosis.

Worked compensation example

A patient has pH 7.20, HCO₃ 12, and PCO₂ 28. This is metabolic acidosis. Winter expected PCO₂ = 1.5(12)+8 = 26 +/-2, so 28 is appropriate. If the measured PCO₂ were 40, the patient would also have respiratory acidosis; if it were 18, the patient would also have respiratory alkalosis.

Figure 4C-2. Anion gap logic



Metabolic acidosis

Metabolic acidosis means the bicarbonate concentration is low enough to drive acidemia unless there is an opposing respiratory process. The most useful first split is high anion gap versus normal anion gap. A high anion gap usually means accumulation of unmeasured organic or inorganic acids. A normal anion gap means bicarbonate has been lost from the body or the kidney cannot excrete acid effectively; chloride rises to balance the fall in bicarbonate, creating hyperchloremic metabolic acidosis.

The anion gap should be interpreted with albumin. Albumin is a major unmeasured anion, so hypoalbuminemia can hide a clinically important gap. A practical correction is to add about 2.5 mEq/L to the calculated anion gap for every 1 g/dL that albumin is below 4 g/dL. For example, an anion gap of 12 with albumin 2.0 corrects to roughly 17, which changes the interpretation.

Formula	How to use it	Clinical trap
Anion gap = $\text{Na} - (\text{Cl} + \text{HCO}_3)$	Normal often about 8-12 mEq/L, but lab ranges vary. Potassium is usually omitted.	A normal-looking gap can be abnormal when albumin is low.
Albumin-corrected AG = $\text{AG} + 2.5 \times (4 - \text{albumin})$	Use when albumin is below normal.	Hypoalbuminemia masks lactic acidosis, ketoacidosis, uremia, or toxins.
Delta ratio = $(\text{AG} - 12) / (24 - \text{HCO}_3)$	Roughly 1-2 suggests isolated high gap acidosis. Below 1 suggests added normal-gap acidosis. Above 2 suggests added metabolic alkalosis or chronic respiratory acidosis.	Delta ratio is a clue, not an absolute rule; use context.
Osmolar gap = measured osmolality - calculated osmolality	Calculated osmolality often = $2\text{Na} + \text{glucose}/18 + \text{BUN}/2.8$, with ethanol/4.6 if present.	Toxic alcohols can have high osmolar gap early and high anion gap later.

High anion gap metabolic acidosis

High anion gap acidosis is caused by accumulation of acids whose conjugate bases are not measured in the standard chemistry panel. Common categories include lactate from tissue hypoperfusion or mitochondrial dysfunction, ketoacids from DKA/alcohol/starvation, uremic acids from kidney failure, and toxins such as methanol, ethylene glycol, propylene glycol, and salicylates. The bedside priority is to determine whether the patient needs fluids, insulin, vasopressors, ventilation, antidote therapy, dialysis, or source control.

Cause	Typical clues	Key initial management link
Lactic acidosis	Shock, sepsis, hypoxemia, seizures, mesenteric ischemia, severe anemia, beta-agonists, metformin in renal failure	Restore perfusion and oxygen delivery; treat sepsis/ischemia; trend lactate after resuscitation.
Diabetic ketoacidosis	Hyperglycemia, ketones, dehydration, abdominal pain, K shifts; total body K depleted despite serum K often high	Fluids, insulin after potassium assessment, potassium replacement, close gap monitoring.
Alcoholic ketoacidosis	Recent binge then poor intake/vomiting; glucose low, normal, or mildly high; high beta-hydroxybutyrate	Dextrose-containing fluids, thiamine, volume and electrolyte repletion.
Starvation ketoacidosis	Prolonged poor intake, pregnancy, eating disorder, illness	Carbohydrate repletion, electrolyte monitoring, consider refeeding risk.
Uremic acidosis	Advanced kidney failure, hyperkalemia, volume overload, uremic symptoms	Treat complications; dialysis if emergent indications.
Methanol	Visual symptoms, CNS depression, high osmolar gap early, high AG later	Fomepizole, bicarbonate if severe acidosis, dialysis for severe cases.
Ethylene glycol	Renal failure, calcium oxalate crystals, hypocalcemia, flank pain, high osmolar gap early	Fomepizole, dialysis in severe poisoning, manage calcium cautiously.
Salicylates	Tinnitus, vomiting, fever, tachypnea; mixed respiratory alkalosis plus metabolic acidosis	Urine alkalinization, activated charcoal when appropriate, dialysis if severe.

Normal anion gap metabolic acidosis

Normal anion gap metabolic acidosis, also called hyperchloremic metabolic acidosis, usually reflects either bicarbonate loss or impaired renal acid excretion. Diarrhea and ileostomy output remove bicarbonate-rich intestinal fluid. Renal tubular acidosis prevents the kidney from acidifying urine, reclaiming bicarbonate, or responding to aldosterone. Large volumes of normal saline can lower bicarbonate by chloride loading and dilutional effects, especially in patients with limited renal reserve.

Mechanism	Examples	Helpful tests
GI bicarbonate loss	Diarrhea, pancreatic or biliary drainage, ileostomy	Volume depletion; low/negative urine anion gap from increased ammonium excretion.
Renal acidification defect	Distal RTA, proximal RTA, type 4 RTA	Urine pH, urine anion gap, potassium pattern, medication and autoimmune review.

Mechanism	Examples	Helpful tests
Medication effect	Acetazolamide, topiramate, amphotericin B, ifosfamide, NSAIDs/ACEi/ARB/heparin causing type 4 physiology	Medication reconciliation; potassium and kidney function.
Iatrogenic chloride load	Large-volume normal saline, chloride-rich resuscitation	Temporal relation to resuscitation; consider balanced crystalloid when appropriate.
Urinary diversion	Ureterosigmoidostomy, ileal conduit	Chloride absorption and bicarbonate loss through bowel segments.

Renal tubular acidosis

Renal tubular acidosis should be suspected when metabolic acidosis is out of proportion to the fall in GFR or when the potassium pattern points toward tubular dysfunction. The urine anion gap is a bedside surrogate for urinary ammonium. In diarrhea, the kidney should excrete ammonium chloride, making the urine anion gap negative. In renal acidification defects, ammonium excretion is inadequate and the urine anion gap is often positive.

RTA type	Core defect	K pattern	Urine pH	Classic associations
Type 1 distal	Cannot secrete H ⁺ in distal nephron	Low K	Often >5.5 despite systemic acidosis	Autoimmune disease, amphotericin B, stones/nephrocalcinosis, hypocitraturia.
Type 2 proximal	Cannot reclaim filtered bicarbonate proximally	Low K	Initially high, then can fall <5.5 after bicarbonate wasting stabilizes	Fanconi syndrome, multiple myeloma, ifosfamide, tenofovir, carbonic anhydrase inhibitors.
Type 4 hypoaldosterone	Low aldosterone effect reduces distal Na reabsorption and H/K secretion	High K	Usually <5.5	Diabetic kidney disease, ACEi/ARB, NSAIDs, heparin, adrenal insufficiency, trimethoprim.

Metabolic alkalosis

Metabolic alkalosis requires both generation and maintenance. Generation usually comes from loss of hydrogen-rich fluid, gain of alkali, or intracellular shifting of hydrogen ions in hypokalemia. Maintenance occurs when the kidney cannot excrete the excess bicarbonate because of volume depletion, chloride depletion, potassium depletion, reduced GFR, or ongoing mineralocorticoid activity. This is why a vomiting patient can stay alkalotic: the stomach generates the alkalosis, but volume/chloride depletion and RAAS activation maintain it.

Type	Urine chloride	Common causes	Treatment logic
Chloride-responsive	Usually <10-20 mEq/L	Vomiting, nasogastric suction, remote diuretic use, post-hypercapnia with volume depletion	Give chloride and volume, usually isotonic saline plus potassium chloride.
Chloride-resistant	Usually >20 mEq/L	Ongoing diuretics, mineralocorticoid excess, Cushing physiology, severe hypokalemia, Bartter/Gitelman patterns	Treat driver: stop diuretic if possible, correct K/Mg, block aldosterone when indicated, evaluate renin/aldosterone.
Alkali load with limited excretion	Variable	Milk-alkali syndrome, bicarbonate/citrate load, renal failure	Stop alkali/calcium sources, restore renal excretion, dialysis if severe kidney failure.

Potassium and alkalosis reinforce each other

Hypokalemia shifts hydrogen ions into cells, increases renal ammoniagenesis, promotes bicarbonate reabsorption, and makes alkalosis harder to correct. In many vomiting or diuretic patients, potassium chloride is not optional supportive care; it is part of the alkalosis treatment.

Respiratory acidosis

Respiratory acidosis means alveolar ventilation is inadequate relative to CO₂ production. The problem may be central drive, neuromuscular transmission, chest wall mechanics, airway obstruction, lung parenchymal disease, or ventilator settings.

Oxygen saturation alone can be misleading because supplemental oxygen may normalize SpO₂ while CO₂ continues to rise. Acute hypercapnia causes acidemia, headache, confusion, asterixis, somnolence, and eventually coma. Chronic hypercapnia is suggested by elevated bicarbonate and a smaller pH disturbance.

Category	Examples	Bedside clues
Low central drive	Opioids, benzodiazepines, alcohol, stroke, brainstem lesion, severe hypothyroidism	Bradypnea, miosis or sedation, low minute ventilation.
Airway obstruction	COPD exacerbation, severe asthma, upper airway obstruction	Wheezing, prolonged expiration, dynamic hyperinflation, rising CO ₂ is ominous in asthma.
Neuromuscular weakness	GBS, myasthenia gravis, ALS, spinal cord injury, botulism	Weak cough, low negative inspiratory force, bulbar symptoms, rapid shallow breathing.
Chest wall/obesity	Obesity hypoventilation, kyphoscoliosis, flail chest	Chronic hypercapnia, sleep-disordered breathing, restrictive mechanics.
Parenchymal/ventilator mismatch	ARDS, pneumonia, ventilator hypoventilation, dead space	High work of breathing or inadequate set minute ventilation.

Management begins with the patient, not the equation. Check mental status, work of breathing, airway protection, hemodynamics, and trajectory. Noninvasive ventilation can be highly effective in COPD exacerbation and cardiogenic pulmonary edema when the patient can protect the airway and cooperate. Intubation is needed for refractory hypoxemia, inability to protect the airway, severe encephalopathy, exhaustion, or worsening acidosis despite noninvasive support. Reverse opioids with naloxone when appropriate, but titrate to restore ventilation rather than to create abrupt withdrawal when possible.

Respiratory alkalosis

Respiratory alkalosis means alveolar ventilation is excessive relative to CO₂ production. The most dangerous mistake is to label it anxiety before ruling out hypoxemia, pulmonary embolism, sepsis, liver failure, pregnancy physiology, salicylates, pain, and CNS disease. Salicylate toxicity is especially important because early direct stimulation of the respiratory center causes respiratory alkalosis, while later organic acids create high anion gap metabolic acidosis. A near-normal pH in salicylate toxicity can hide a severe mixed disorder.

Cause group	Examples	Diagnostic clue
Hypoxemic drive	Pulmonary embolism, pneumonia, high altitude, severe anemia	Low PaO ₂ or increased A-a gradient; tachypnea out of proportion.
Systemic illness	Sepsis, fever, pain, pregnancy, liver failure	Respiratory alkalosis may precede hypotension in sepsis.
CNS/medication	Stroke, intracranial pathology, salicylates, progesterone	Tinnitus and mixed disorder suggest salicylates.
Iatrogenic	Excessive ventilator minute ventilation	Low PCO ₂ after ventilator changes; adjust settings to target physiology.
Psychogenic hyperventilation	Panic, anxiety	Diagnosis of exclusion; paresthesias and carpopedal spasm can occur from alkalemia lowering ionized calcium.

Mixed disorders

Mixed acid-base disorders are common in hospitalized patients. A septic patient with renal failure may have lactic acidosis plus uremic acidosis. A vomiting patient with DKA may have high anion gap acidosis plus metabolic alkalosis, making the bicarbonate look less low than expected. A COPD patient on diuretics may have chronic respiratory acidosis plus metabolic alkalosis. The most reliable strategy is to check compensation, calculate the gap, correct for albumin, and ask whether the delta relationship fits the bicarbonate change.

Pattern	What it suggests
pH near normal but PCO ₂ and HCO ₃ both very abnormal	Mixed or chronic compensated disorder. Do not call it normal.

Pattern	What it suggests
Metabolic acidosis with PCO ₂ higher than Winter expected	Added respiratory acidosis: fatigue, COPD, sedatives, neuromuscular weakness.
Metabolic acidosis with PCO ₂ lower than Winter expected	Added respiratory alkalosis: sepsis, pregnancy, salicylates, liver disease, PE.
High anion gap with delta ratio <1	Added normal-gap metabolic acidosis: diarrhea, RTA, saline.
High anion gap with delta ratio >2	Added metabolic alkalosis or chronic respiratory acidosis.
Alkalemia with elevated anion gap	Metabolic alkalosis masking high-gap acidosis; consider vomiting plus DKA/toxin/lactate.

Osmolar gap and toxic alcohols

The osmolar gap compares measured serum osmolality to calculated osmolality. A high osmolar gap means there are osmotically active solutes not included in the calculation, such as ethanol, methanol, ethylene glycol, isopropanol, propylene glycol, mannitol, or radiocontrast. In toxic alcohol poisoning, the osmolar gap is often highest early when parent alcohol is present; the anion gap rises later as toxic acidic metabolites accumulate. A normal osmolar gap does not fully exclude a late toxic alcohol ingestion if the clinical story is strong.

Substance	Acid-base pattern	Signature clues	Treatment anchor
Methanol	High osmolar gap early, high AG acidosis later	Visual blurring, "snowfield" vision, basal ganglia injury	Fomepizole, folinic/folic acid, bicarbonate, dialysis when severe.
Ethylene glycol	High osmolar gap early, high AG acidosis later	Calcium oxalate crystals, renal failure, hypocalcemia	Fomepizole, thiamine/pyridoxine sometimes used, dialysis when severe.
Isopropanol	High osmolar gap with ketosis but usually no high AG acidosis	Hemorrhagic gastritis, acetone odor, CNS depression	Supportive care; dialysis rarely for severe cases.
Propylene glycol	High osmolar gap and lactic/high AG acidosis possible	IV lorazepam or medication solvent exposure, ICU context	Stop source, supportive care, dialysis if severe.
Salicylates	Mixed respiratory alkalosis and high AG metabolic acidosis	Tinnitus, vomiting, fever, tachypnea	Activated charcoal, bicarbonate/urine alkalinization, dialysis when severe.

Diagnostic approach by presentation

Acid-base results should be integrated with volume status, potassium, chloride, renal function, glucose, ketones, lactate, osmolality, medications, and urine studies. In metabolic acidosis, always ask whether the patient is producing acids, failing to excrete acids, or losing bicarbonate. In metabolic alkalosis, ask whether urine chloride is low because the kidney wants chloride, or high because the driver is ongoing diuretics/mineralocorticoid physiology. In respiratory disorders, ask whether the problem is ventilation, oxygenation, or both.

Initial data cluster	Why it matters
BMP with Na, K, Cl, HCO ₃ , BUN, creatinine	Calculates anion gap, finds K danger, assesses kidney contribution.
ABG or VBG plus lactate	Names primary process and identifies shock/ischemia physiology.
Serum beta-hydroxybutyrate or ketones	Detects DKA, alcoholic ketoacidosis, starvation ketoacidosis.
Measured serum osmolality and ethanol level	Calculates osmolar gap and avoids false toxic alcohol interpretation.
Urinalysis and urine electrolytes	RTA clues, ketones, crystals, urine pH, urine anion gap.
Medication/toxin review	Diuretics, acetazolamide/topiramate, ACEi/ARB, NSAIDs, salicylates, toxic alcohols, metformin.
ECG	Hyperkalemia/hypokalemia effects may be immediately life-threatening.

Initial management principles

Treat the patient and the cause. A pH abnormality is a marker of physiology, not the sole therapeutic target. Airway and ventilation come first in respiratory failure. Perfusion, oxygen delivery, and source control come first in lactic acidosis. Fluids, insulin, and potassium-aware protocols come first in DKA. Antidotes and dialysis evaluation come first in suspected toxic alcohols. Chloride and potassium repletion come first in many metabolic alkaloses. Bicarbonate therapy is situation-specific and can worsen sodium load, volume status, CO₂ generation, ionized calcium, and intracellular acidosis if used indiscriminately.

Emergency	Immediate actions
Severe hyperkalemia with ECG changes	Calcium for membrane stabilization, insulin/glucose, beta-agonist, bicarbonate when acidemic, remove K with diuretics/binders/dialysis as appropriate.
Severe metabolic acidosis with shock	Resuscitate perfusion, oxygenation, infection/ischemia control; consider bicarbonate selectively in severe acidemia, especially with kidney failure or hyperkalemia.
DKA	Isotonic fluid resuscitation, potassium assessment before insulin if K is low, insulin infusion, dextrose when glucose falls, close gap monitoring.
Toxic alcohol suspicion	Do not wait for levels if high suspicion: call poison control, give fomepizole, check osmolar/anion gaps, consult nephrology for dialysis criteria.
Metabolic alkalosis with volume depletion	Normal saline and potassium chloride, stop vomiting/NG losses if possible, hold diuretics when safe.
Acute ventilatory failure	Reverse cause if possible, consider NIV when appropriate, intubate for airway failure, exhaustion, refractory acidosis, or inability to protect airway.

Common pitfall

Do not chase a normal pH. A patient with pH 7.40, PCO₂ 20, HCO₃ 12, and anion gap 24 has a major high-gap metabolic acidosis plus respiratory alkalosis, not a normal acid-base state.

High-yield exam clues

- DKA: high anion gap acidosis, ketones, dehydration, total body potassium deficit even when serum potassium is high.
- Salicylate toxicity: respiratory alkalosis plus high anion gap metabolic acidosis; tinnitus and tachypnea are classic clues.
- Ethylene glycol: high anion gap acidosis, high osmolar gap early, calcium oxalate crystals, renal failure, hypocalcemia.
- Methanol: high anion gap acidosis plus visual symptoms; osmolar gap may be absent late.
- Diarrhea: normal anion gap metabolic acidosis with hypokalemia and usually negative urine anion gap.
- Type 1 distal RTA: urine pH stays inappropriately high during systemic acidosis; kidney stones/nephrocalcinosis.
- Type 4 RTA: hyperkalemic normal-gap acidosis, often in diabetic kidney disease or medication-related hypoaldosterone states.
- Vomiting/NG suction: metabolic alkalosis, hypochloremia, hypokalemia, low urine chloride unless ongoing diuretics or mineralocorticoid excess.
- COPD: chronic respiratory acidosis has elevated bicarbonate; acute worsening lowers pH and raises PCO₂ above baseline.
- Pregnancy, PE, sepsis, liver failure, and salicylates can all cause respiratory alkalosis.

Practice questions

1. A 23-year-old with type 1 diabetes has vomiting, glucose 510, Na 132, Cl 94, HCO₃ 8, pH 7.12, PCO₂ 23. What is the primary disorder and is compensation appropriate?

Answer: High anion gap metabolic acidosis from DKA. Winter expected PCO₂ = 1.5(8)+8 = 20 +/-2; measured 23 is slightly higher than expected, suggesting mild added respiratory acidosis or fatigue depending on lab tolerance and clinical status.

2. A patient with profuse diarrhea has Na 140, Cl 114, HCO₃ 14, pH 7.28, PCO₂ 30. What type of metabolic acidosis is likely?

Answer: Normal anion gap metabolic acidosis from GI bicarbonate loss. The kidney should increase ammonium excretion, so urine anion gap is often negative.

3. A patient with COPD has pH 7.36, PCO₂ 62, HCO₃ 34. How should this be interpreted?

Answer: Chronic respiratory acidosis with renal compensation. The near-normal pH is not normal physiology; it reflects chronic compensation.

4. A vomiting patient has pH 7.53, PCO₂ 48, HCO₃ 39, urine chloride 6 mEq/L. What treatment matches the mechanism?

Answer: Chloride-responsive metabolic alkalosis. Treat with isotonic saline and potassium chloride while addressing vomiting/NG losses.

5. A patient has pH 7.41, PCO₂ 19, HCO₃ 12, anion gap 25, tinnitus, nausea, and fever. What is the likely diagnosis?

Answer: Salicylate toxicity causing mixed respiratory alkalosis and high anion gap metabolic acidosis.

6. A patient after antifreeze ingestion has high osmolar gap, high anion gap acidosis, hypocalcemia, and AKI. What is the toxin and management anchor?

Answer: Ethylene glycol. Give fomepizole promptly, involve poison control/nephrology, and consider dialysis in severe poisoning.

7. A diabetic patient on lisinopril and trimethoprim has K 6.0, HCO₃ 17, normal anion gap, and urine pH 5.2. What RTA pattern fits?

Answer: Type 4 RTA from hypoaldosterone effect/resistance, producing hyperkalemic normal-gap metabolic acidosis.

8. A septic patient has pH 7.24, HCO₃ 12, PCO₂ 18. Winter expected PCO₂ is about 26 +/-2. What mixed process exists?

Answer: Metabolic acidosis plus respiratory alkalosis. In sepsis, hyperventilation can lower PCO₂ more than expected.

Graph-RAG extraction map

The purpose of this map is to make the chapter easy to convert into nodes and relationships for Medvale graph retrieval. Each row can become a concept node, edge template, or retrieval anchor.

Node / concept	Class memberships	Key edges to activate
Acid-base disorder	renal, electrolyte, physiology, diagnostic framework	has-primary-variable: pH/PCO ₂ /HCO ₃ ; evaluated-by: ABG/VBG; modifies: potassium and ionized calcium.
Metabolic acidosis	acid-base, emergency, renal	subdivides-into: high anion gap and normal anion gap; compensated-by: respiratory alkalosis.
Anion gap	calculation, diagnostic discriminator	increased-by: lactate/ketones/uremia/toxins; masked-by: hypoalbuminemia.
Delta ratio	mixed disorder tool	detects: additional normal-gap acidosis or metabolic alkalosis.
Osmolar gap	toxicology, calculation	suggests: toxic alcohols, ethanol, propylene glycol, mannitol; changes-over-time with metabolism.
Renal tubular acidosis	tubular disease, normal-gap acidosis	typed-by: distal/proximal/type 4; differentiated-by: potassium and urine pH.
Metabolic alkalosis	acid-base, volume, chloride	maintained-by: volume/chloride/K depletion or mineralocorticoid excess; treated-by: chloride/K repletion when responsive.
Respiratory acidosis	pulmonary, ventilation	caused-by: hypoventilation; acute-vs-chronic by bicarbonate response; treated-by: ventilation support.
Respiratory alkalosis	pulmonary, systemic illness	caused-by: hyperventilation; associated-with: PE/sepsis/pregnancy/salicylates.
Toxic alcohol poisoning	toxicology, emergency	causes: osmolar gap then anion gap; treated-by: fomepizole/dialysis depending severity.

References and source notes

Uploaded renal/fluids source PDF: acid-base coverage signal from pages 42-54, including metabolic acidosis, anion gap diagrams, metabolic alkalosis, respiratory alkalosis, respiratory acidosis, and treatment themes. Anki CSV used as a topic refresher only; the chapter text, tables, and diagrams are original Medvale synthesis.

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